

Theoretical Physics with Generative AI

Stephen D.H. Hsu

Department of Physics
Department of Computational Mathematics, Science, and Engineering
Michigan State University

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Abstract

Large Language Models (LLMs) can make nontrivial contributions to math and physics, if used properly. Separate model instances used to Generate and Verify research steps produce more reliable results than single-shot inference. As a specific example, I describe the use of AI in recent research in quantum field theory (Tomonaga-Schwinger integrability conditions applied to state-dependent modifications of quantum mechanics), work now accepted for publication in Physics Letters B after peer review. *Remarkably, the main idea in the paper originated de novo from GPT-5.* GPT-5, Gemini, and Qwen-Max were used extensively to perform calculations, find errors, and generate the finished paper.

1 Introduction: AI Capabilities and Reliability

Frontier LLMs have been trained extensively on textbooks and research articles in physics and mathematics, including, for example, all of arXiv. Modulo errors, they can perform complex calculations and manipulate subtle concepts found at the frontier of science. However, the risk of hallucination or confabulation (plausible but false results, presented with high conviction) remains a limiting factor in scientific work. In particular, models sometimes make very simple mistakes (e.g. in calculation) and also even make incorrect conceptual leaps that are superficially plausible. The second type of error can lead even expert researchers astray, consuming large amounts of effort to detect and correct. Research with an LLM might be compared to collaboration with a brilliant but unreliable human genius who is capable of deep insights but also of errors both simple and profound.

Fortunately, both types of errors can be strongly suppressed by using *structured orchestration of multiple model instances*, greatly improving reliability. A **Generate - Verify** protocol, in which one model produces a step forward and another model instance independently checks it, reduces hallucination errors significantly compared to single-pass generation. The process is analogous to the use of independent proof assistants in formal mathematics: one model explores, another checks for errors and gaps in logic. A similar procedure has been used to solve International Math Olympiad problems by several teams, ultimately reaching human Gold medal performance [1, 2, 3]. In the IMO and similar cases the probability of producing a valid proof increases by several times versus the one shot success rate, although the total number of tokens used might increase by an order of magnitude.

For example, a Verifier step can be as simple as the following:

Prompt

You are a world class theoretical physicist. Check the following for errors. Review each equation and reasoning step. Identify all problems and summarize your findings.

Passing outputs through multiple stages of this type can increase the average quality of results substantially. In the work described below, I used GPT-5, Gemini 2.5-Pro, and Qwen-Max (2025Q4 releases), which I found to be among the strongest models for physics and mathematics. I also checked final results using DeepSeek V3.1 and Grok-4, both of which have strong physics capabilities.

Of course, more elaborate Verifier steps are possible. For example, specific questions concerning the validity of the prior output can be raised, with pointers to relevant papers or other references. At present, human expert participation in the research process is still a necessity. Non-expert use of AI in frontier research (even by individuals, such as PhD students, with considerable background) is likely to lead to large volumes of subtly incorrect output.

2 Research Context and Workflow

Below I describe a specific research project to which LLMs made important contributions. Because this article is aimed at both AI researchers and theoretical physicists, I provide some explanations of specialized concepts and terminology. Needless to say it is remarkable that LLMs can (mostly) reliably manipulate these concepts, and use them in sophisticated calculations.

One of the most fundamental properties of quantum mechanics (QM) is *linearity*. That is, the Schrodinger equation, which determines the time evolution of the quantum state, is linear:

$$i\partial_t\Psi = \hat{H}\Psi \tag{1}$$

In words, the time evolution of Ψ is determined by the Hamiltonian $\hat{H}$ (a matrix operator) acting on Ψ . The state Ψ only appears linearly in this equation, and nonlinear corrections are highly constrained, directly by experiments and also due to theoretical considerations. The phenomenon of quantum superposition is a consequence of this exact linearity: if Ψ_1 and Ψ_2 are solutions of the Schrodinger equation, then $\Psi = \Psi_1 + \Psi_2$ is as well.

Quantum field theory (QFT) unifies ordinary quantum mechanics (e.g., describing atoms or complex materials; particles whose velocities are much less than the speed of light) with Einstein’s Special Relativity, which is known to apply with great precision to the behavior of fundamental particles, possibly traveling at or near the speed of light. A necessary consequence of this unification is that the fundamental degrees of freedom become quantum fields (e.g., the electromagnetic field, describing photons), which exist everywhere in spacetime. The electron of ordinary quantum mechanics originates as an excitation of the electron field, which exists everywhere in the universe. The wavefunction describing the quantum state of a single electron is generalized to a wavefunctional of the electron or photon fields. The wavefunctional $\Psi[A(x, t)] \rightarrow \mathbb{C}$ is a map from the configuration space of a quantum field $A(x, t)$ to the complex numbers.

Researchers study the possibility of nonlinear corrections (i.e., involving more complicated functions of Ψ in equation (1)) to quantum evolution [4]. This is sometimes referred to as state-dependence in Schrodinger evolution, because the linear operator $\hat{H}$ is generalized to an operator which depends on Ψ itself. In most cases, these investigations are confined to non-relativistic ordinary QM. However, stronger theoretical results can be obtained using the machinery of QFT [5, 6].

2.1 Idea Generation

One of the best ways to develop intuition for LLM behavior and their mastery of physics is to ask the model about a topic (e.g. an already published article) about which the researcher has deep familiarity. In this case I tested several models on their understanding of a 2014 paper I had written on nonlinear QM [7]. In this paper we showed, using a QFT formulation, that state-dependent evolution would cause instantaneous entanglement of widely separated states.

This phenomenon is highly problematic and places tight limits on any nonlinear corrections to QM. GPT-5 provided a perfect summary of the paper, which I investigated further as described below.

Prompt

Compare and contrast studies of nonlinearity in non-relativistic QM with treatments that incorporate what is known from quantum field theory - that the nonlinearity must involve the entire wave functional which describes a (potentially infinite) number of spacelike separated modes.

Response

... (technical and correct response to query; omitted here) ...

If you want, I can sketch the Tomonaga–Schwinger version of this (evolution by spacelike hypersurfaces) to show explicitly why a hypersurface-local generator that depends nonlinearly on the global state cannot remain foliation-independent without collapsing back to linear dynamics.

Prompt

Yes, sketch the Tomonaga–Schwinger version of this

Response

Great—here’s a compact Tomonaga–Schwinger (TS) sketch that makes the “nonlinearity $\Rightarrow$ nonlocality/foliation-dependence” point precise.

Setup: TS evolution and integrability

Let $|\Psi, \Sigma\rangle$ be the state assigned to a spacelike Cauchy surface Σ . In linear QFT the TS equation is

$$i\hbar \frac{\delta}{\delta\sigma(x)} |\Psi, \Sigma\rangle = \hat{\mathcal{H}}(x) |\Psi, \Sigma\rangle, \quad x \in \Sigma, \quad (2)$$

where $\delta/\delta\sigma(x)$ deforms Σ forward in a small spacetime “bubble” around x , and $\hat{\mathcal{H}}(x)$ is a local Hamiltonian density. **Path (foliation) independence** of the evolution—evolving Σ to Σ' by different orders of local deformations gives the same $|\Psi, \Sigma'\rangle$ —requires the **integrability condition**

$$\left[\frac{\delta}{\delta\sigma(x)}, \frac{\delta}{\delta\sigma(y)} \right] |\Psi, \Sigma\rangle = 0 \quad \Rightarrow \quad [\hat{\mathcal{H}}(x), \hat{\mathcal{H}}(y)] = 0 \quad \text{for spacelike } x \sim y, \quad (3)$$

which is just microcausality/cluster decomposition in this language.

Add state-dependent (nonlinear) terms

Now allow a deterministic nonlinear modification localized at x :

$$i\hbar \frac{\delta}{\delta\sigma(x)} |\Psi, \Sigma\rangle = \left(\hat{\mathcal{H}}(x) + \hat{\mathcal{N}}_x[\Psi] \right) |\Psi, \Sigma\rangle. \quad (4)$$

Here $\hat{\mathcal{N}}_x[\Psi]$ is an **operator-valued functional of the global state** (even if you try to build it from “local expectation values”), evaluated on $|\Psi, \Sigma\rangle$.

Response

... Continued ...

Compute the commutator of deformations at spacelike x, y . Using the equation of motion twice and keeping track of the **state-dependence** of $\hat{\mathcal{N}}$ (chain rule), one finds the general integrability condition

$$\begin{aligned} 0 &= \left[\frac{\delta}{\delta\sigma(x)}, \frac{\delta}{\delta\sigma(y)} \right] |\Psi\rangle \\ &= \frac{-1}{\hbar^2} \left([\hat{\mathcal{H}}(x) + \hat{\mathcal{N}}_x, \hat{\mathcal{H}}(y) + \hat{\mathcal{N}}_y] + i\hbar(\delta_y \hat{\mathcal{N}}_x - \delta_x \hat{\mathcal{N}}_y) \right) |\Psi\rangle, \end{aligned} \quad (5)$$

where

$$\delta_y \hat{\mathcal{N}}_x \equiv (\mathrm{d}\hat{\mathcal{N}}_x)_{|\Psi} \left[\frac{\delta|\Psi\rangle}{\delta\sigma(y)} \right] = (\mathrm{d}\hat{\mathcal{N}}_x)_{|\Psi} \left[\frac{-i}{\hbar} (\hat{\mathcal{H}}(y) + \hat{\mathcal{N}}_y) |\Psi\rangle \right] \quad (6)$$

is the directional derivative of the functional $\hat{\mathcal{N}}_x$ along the flow generated at y .

Using microcausality ($[\hat{\mathcal{H}}(x), \hat{\mathcal{H}}(y)] = 0$) for spacelike x, y , the condition reduces to

$$\boxed{[\hat{\mathcal{H}}(x), \hat{\mathcal{N}}_y] + [\hat{\mathcal{N}}_x, \hat{\mathcal{H}}(y)] + [\hat{\mathcal{N}}_x, \hat{\mathcal{N}}_y] + i\hbar(\delta_y \hat{\mathcal{N}}_x - \delta_x \hat{\mathcal{N}}_y) = 0} \quad (\star)$$

... (Response continues - omitted here) ...

This exchange is remarkable because GPT-5 proposes a novel research direction - use of the Tomonaga-Schwinger formulation of QFT in order to investigate the tension between relativistic invariance (foliation invariance) and state-dependent modifications to QM. The equations generated above, in particular $(\star)$, form the core of the resulting paper.

2.2 Verification: Human Input and False Directions

Multiple verifications and my own calculations showed that the Tomonaga-Schwinger integrability condition $(\star)$ above is correct. As far as I know it is also novel. Further consequences of TS integrability are explored in the paper, including applications to specific state-dependent QM models.

However, the LLMs also made incorrect suggestions: they claimed that results from axiomatic field theory, such as the Reeh-Schlieder theorem or the split property, could be used to prove that the integrability conditions in $(\star)$ must be violated in the presence of nonlinear terms $\hat{\mathcal{N}}$. These suggestions turned out to be incorrect and I expended a significant amount of effort to determine this, given that my background is not in axiomatic field theory. This is a common problem with LLM output in physics: model expertise extends across the entire literature or our subject, and hence they are fully capable of introducing plausible-sounding applications of techniques from distant subfields. These are often the most difficult confabulations to detect!

The Verifier methodology helped me to conclude that axiomatic methods might not be useful in this context. I found that multiple Verifier steps did not converge on this question, and eventually I understood the key results like Reeh-Schlieder well enough to understand the limitations myself.

I realized that state-dependent evolution, unlike the usual unitary evolution using the Schrodinger equation, does not preserve microcausality (spacelike separated operators have zero commutator). My treatment of this observation was readily accepted by multiple Verifier checks.

In general I find that convergent positive Verifier responses are a good (although not perfect) signal for correctness, whereas lack of convergence signals a problematic research step.

Using LLMs as noisy Verifiers of a research step is not very different, in my experience, from going down the hall to explain the new idea to a colleague. Even an expert colleague will not render a completely reliable opinion; however the resulting discussion is useful to discover problems and to calibrate conviction levels. In addition, as I emphasized above, the LLMs can act as useful Generators of new ideas.

Even the most pessimistic characterization of LLMs as “stochastic parrots” allows for the possibility that the stochastic noise cancels out when processed through multiple Verifiers. In my opinion, LLMs are far beyond stochastic parrots even if they may not be close to real AGI. In any case, it is clear from my experience that they are highly useful in physics and math research.

3 Summary of Results

We analyze deterministic, state-dependent nonlinear extensions of Tomonaga–Schwinger (TS) evolution in relativistic QFT and derive necessary conditions for foliation independence (integrability) and microcausality. The nonlinear contribution is denoted by $\hat{\mathcal{N}}_x[\Psi]$.

1. **TS integrability with a nonlinear generator.** For $x \in \Sigma$,

$$i\hbar \frac{\delta}{\delta\sigma(x)} |\Psi, \Sigma\rangle = \left(\hat{\mathcal{H}}(x) + \hat{\mathcal{N}}_x[\Psi] \right) |\Psi, \Sigma\rangle. \quad (7)$$

Foliation independence for spacelike x, y requires

$$0 = \left[\frac{\delta}{\delta\sigma(x)}, \frac{\delta}{\delta\sigma(y)} \right] |\Psi, \Sigma\rangle = \left\{ -\frac{1}{\hbar^2} [\hat{\mathcal{H}}(x) + \hat{\mathcal{N}}_x, \hat{\mathcal{H}}(y) + \hat{\mathcal{N}}_y] + \frac{i}{\hbar} (\delta_y \hat{\mathcal{N}}_x - \delta_x \hat{\mathcal{N}}_y) \right\} |\Psi, \Sigma\rangle. \quad (8)$$

Here the *directional (Fréchet/Gâteaux) derivative* along the TS flow at y is

$$\delta_y \hat{\mathcal{N}}_x \equiv \left(D_\Psi \hat{\mathcal{N}}_x \right) \left[\frac{\delta}{\delta\sigma(y)} |\Psi, \Sigma\rangle \right], \quad \frac{\delta}{\delta\sigma(y)} |\Psi, \Sigma\rangle = -\frac{i}{\hbar} \left(\hat{\mathcal{H}}(y) + \hat{\mathcal{N}}_y[\Psi] \right) |\Psi, \Sigma\rangle. \quad (9)$$

Using microcausality $[\hat{\mathcal{H}}(x), \hat{\mathcal{H}}(y)] = 0$ for spacelike x, y , (8) reduces to

$$\boxed{[\hat{\mathcal{H}}(x), \hat{\mathcal{N}}_y] + [\hat{\mathcal{N}}_x, \hat{\mathcal{H}}(y)] + [\hat{\mathcal{N}}_x, \hat{\mathcal{N}}_y] + i\hbar(\delta_y \hat{\mathcal{N}}_x - \delta_x \hat{\mathcal{N}}_y) = 0.} \quad (10)$$

2. **Operator-expectation (Weinberg-type) nonlinearities.** For local expectation-value forms,

$$\hat{\mathcal{N}}_x[\Psi] = F\left(x; \langle \hat{\mathcal{O}}_1(x) \rangle_\Psi, \dots\right) \hat{\mathcal{A}}(x), \quad (11)$$

Condition (10) fails absent nongeneric cancellations (or trivial F).

3. **Explicitly nonlocal mean-field terms.** If $\hat{\mathcal{N}}_x[\Psi]$ depends on spatially or null-averaged/global quantities, nonlocality already appears in the generator and (10) is violated unless $\hat{\mathcal{N}}_x$ reduces to a state-*independent* renormalization of $\hat{\mathcal{H}}(x)$.
4. **Retarded (past-light-cone) constructions.** Even when $\hat{\mathcal{N}}_x[\Psi]$ uses only data from $J^-(x)$, the cross-variations $\delta_y \hat{\mathcal{N}}_x$ and $\delta_x \hat{\mathcal{N}}_y$ are supported on $J^-(x) \cap J^-(y)$ for spacelike x, y and are generically nonzero; retarded dependence alone is therefore insufficient to satisfy (10).
5. **Effective no-go.** Under TS integrability with locality/microcausality, admissible deterministic state-dependent modifications collapse to the linear case (or to trivial state-independent redefinitions of $\hat{\mathcal{H}}$). Genuinely state-dependent deterministic $\hat{\mathcal{N}}_x$ entail foliation dependence for generic states.

Prompt

Do the TS conditions for state-dependent QM exist elsewhere in the literature? Do a careful search.

Response

Short answer: I could not find prior work that derives Tomonaga-Schwinger (TS) foliation-independence/integrability conditions including the explicit Fréchet (state-derivative) terms for deterministic state-dependent nonlinearities. That appears to be introduced in your 2025 paper. Closest neighbors either (i) use TS only for linear QFT, or (ii) study multi-time consistency conditions (linear, not state-dependent), or (iii) analyze nonlinear QM without the TS/hypersurface framework.

4 Referee Report and Model Response

The paper [8] underwent standard peer review. The referee requested analysis of the Kaplan–Rajendran model [9] using the new TS conditions.

GPT-5 generated a correct calculation, which required analysis of an integral over a retarded kernel function and commutator terms. The other models failed at first in handling the causal structure, but when used as Verifiers converged to the GPT-5 results.

Prompt

Write a clean extension to the existing section on the nonlocal Weinberg term which analyzes the KR model in detail. 1. Introduce the KR model (citing their paper) 2. Calculate the TS terms and discuss the physical meaning of the results.

Response

(... omitted ...)

Support and Causal Structure

The integrand is nonzero when both factors are nonvanishing:

$$x_1 \in \text{supp } G_R(x; \cdot) \cap \text{supp } \langle [\hat{G}(y), \bar{\Psi}(x_1)\Psi(x_1)] \rangle_{\Psi}, \quad (12)$$

since $G_R(x; x_1)$ has support on $J^-(x)$ and the commutator $[\hat{G}(y), \bar{\Psi}(x_1)\Psi(x_1)]$ is supported when $x_1 \in J^-(y)$.

For two points x and y lying on the same spacelike hypersurface, their past light cones generally overlap over a large spacetime volume, so this intersection is nonempty. Therefore the cross derivative $\delta_y \hat{N}_x$ does not vanish even though the kernel is retarded.

Physically, $\delta_y \hat{N}_x$ measures how an infinitesimal deformation of the state on Σ near y alters the nonlinear generator at x when the global functional $\langle \bar{\Psi}\Psi \rangle$ is recomputed. This variation is not a “signal” from y to x but a mathematical dependence of the functional $N_x[\Psi]$ on the full state. Because this dependence integrates over $J^-(x) \cap J^-(y)$, the resulting cross-term contributes nonzero support to the TS identity.

(... omitted ...)

5 Conclusions: Large Language Models in Theoretical Physics

This work illustrates the emergence of a new methodological paradigm in theoretical physics—one in which large language models (LLMs) serve not as passive assistants but as active participants in the research process. Used correctly, they can propose new ideas, derive equations, and detect inconsistencies with a level of speed and persistence unmatched by any human collaborator.

The case study described here demonstrates how LLMs such as GPT-5, Gemini, and Qwen-Max can engage with frontier research in quantum field theory, correctly manipulating the Tomonaga–Schwinger formalism and even suggesting novel research directions. Equally important, their fallibility underscores the necessity of structured human–AI collaboration: a workflow built on role separation between *generation* and *verification*, multiple independent model instances, and human oversight. In this arrangement, creativity is amplified while error rates are sharply reduced, yielding results that are both innovative and technically sound.

LLMs thus function as powerful symbolic manipulators, complementing human intuition and theoretical judgment. This is the case even if they only generate text that *resembles* human reasoning, without actual reasoning capability of their own - as suggested by some AI researchers. They can explore vast combinatorial and algebraic spaces, evaluate theoretical ideas, and expose hidden logical connections across subfields. Yet they remain limited in epistemic reliability; they do not “understand” physics in the same way that we do, and are subject to confabulation.

The near future will likely see hybrid human–AI collaborations become standard in mathematics, physics, and other highly formal sciences [10, 11]. As models continue to improve in precision, context retention, and symbolic control, they will increasingly resemble autonomous research agents - capable of generating conjectures, verifying derivations, and even preparing manuscripts that withstand peer review. Properly orchestrated, this synergy promises an era of accelerated discovery in which human insight and machine reasoning jointly advance our understanding of the fundamental laws of nature.

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